To explain these important aspects, let's first call upon an example that pares away extraneous details without sacrificing the new physics. Instead of considering all ten of the spacetime dimensions of string theory—or even the four extended spacetime dimensions we are familiar with—let's go back to the Garden-hose universe. We originally introduced this two-spatial-dimension universe in Chapter 8 in a prestring context to explain aspects of Kaluza's and Klein's insights in the 1920s. Let's now use it as a "cosmological playground" to explore the properties of string theory in a simple setting; we will shortly use the insights we gain to better understand all of the spatial dimensions string theory requires. Toward this end, we imagine that the circular dimension of the Garden-hose universe starts out nice and plump but then shrinks to shorter and shorter size, approaching the form of Lineland—a simplified, partial version of the big crunch.

The question we seek to answer is whether the geometrical and physical properties of this cosmic collapse have features that markedly differ between a universe based on strings and one based on point particles.

The Essential New Feature

We do not have to search far to find the essential new string physics. A point particle moving in this two-dimensional universe can execute the kinds of motion illustrated in Figure 10.2: It can move along the extended dimension of the Garden-hose, it can move along the curled-up part of the Garden-hose, or any combination of the two. A loop of string can undergo similar motion, with one difference being that it oscillates as it moves around on the surface, as shown in Figure 10.3(a). This is a distinction we have already discussed in some detail: The oscillations of the string imbue it with characteristics such as mass and force charges. Although a crucial aspect of string theory, this is not our present focus, since we already understand its physical implications.

Instead, our present interest is in another difference between point-particle and string motion, a difference directly dependent on the shape of the space through which the string is moving. Since the string is an extended object, there is another possible configuration beyond those already mentioned: It can wrap around—lasso, so to speak—the circular part of the Garden-hose universe, as shown in Figure 10.3(b).1